

Nikhil Ashok

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SUMMARY

Results-driven Computer Science graduate specializing in embedded systems hardware, IoT, Flutter application deployment, and smart product delivery. Experienced in deploying scalable IoT products alongside startup teams, cutting solution costs by 50%, and building cloud-connected hardware pipelines using Firebase. Proven track record of integrating practical ML models and leading high-impact technical events.

TECHNICAL SKILLS

- **Programming Languages:** Dart, C, C++, JavaScript, Python, Assembly
- **Tools & Technologies:** Git, GitHub, VS Code, Android Studio, Postman, Streamlit
- **Frameworks:** Flutter
- **IoT & Hardware:** ESP32, ESP32-S3, Raspberry Pi 3B, Arduino Uno, NodeMCU ESP8266, Sensor Interfacing & Calibration
- **AI & Machine Learning:** Gemini API, Artificial Neural Networks (ANN), Random Forest, XGBoost, Model Training & Evaluation
- **Cloud & Backend:** Google Firebase (Realtime Database, Firestore, Auth, Cloud Functions), Supabase, REST APIs, SQL

WORK EXPERIENCE

IoT Products Developer (Intern)

June 2025 – December 2025

AsTech Solutions

- Engineered end-to-end IoT solutions by integrating microcontrollers (ESP32) with cloud databases and Flutter applications, enabling real-time remote monitoring and device control.
- Developed and deployed an automated Smart Sanitary Pad Dispenser with biometric authentication for NIC Peshawar, servicing 200+ active facility users with a 99% dispensing success rate.
- Designed a Universal Multi-Frequency RFID Reader replacing 3 standalone units, cutting hardware production costs by 50% and streamlining multi-card scanning efficiency.
- Interfaced IR sensor modules with ESP32 microcontrollers to retrofit conventional AC units into smart appliances, enabling automated remote controlling, scheduling and power optimization.

EDUCATION

Bachelor of Science in Computer Science

October 2022 – August 2026

CECOS University of IT and Emerging Sciences, Peshawar

Grade / CGPA: 4.0/4.0

- **Relevant Coursework:** Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), IoT & Embedded Systems, Database Systems, Mobile App Development, Web Engineering, Information Security, Game Development, Cloud Computing, Computer Networks, Parallel and Distributed Computing, Artificial Intelligence, Machine Learning, Operating Systems, Algorithm Analysis.

PROJECTS

Aqua Sentinel (IoT & ML-Based Water Management System)

- Developed an ESP32-based smart water management prototype utilizing flow and quality sensors to automate leak detection, water quality monitoring, and usage tracking via a custom smart meter with real-time alerts, consumption analytics, and automated water pump control in a mobile app.
- Integrated the Gemini API into a cross-platform Flutter application to deliver personalized water conservation tips and AI-driven telemetry analytics for users.

IoT Sanitary Napkins Dispenser (Female Health Tech)

- Designed an automated pad dispensing mechanism equipped with biometric access using standalone fingerprint modules and ESP32 hardware.
- Interfaced ESP32 hardware with a full-stack Flutter mobile application via Bluetooth to enable remote biometric profile registration and manage admin controls.

CERTIFICATIONS

Soft Skills Training Program

June 2026

Pakistan Software Export Board (PSEB)

- **Focus:** Strategic Leadership, Problem Solving, Critical Thinking, Presentation Skills, Career Readiness, Business Writing, Client Negotiation, Executive Communication, and Key Account Management.

The Complete Flutter Development Bootcamp with Dart

2025

Udemy

Credential ID: UC-a1505838-0a61-46c8-9fc7-02c20e06367a

- **Skills:** Cross-platform mobile app development, state management, REST API integration, Core Flutter & Dart Concepts, UI & Design Skills, and full-stack Flutter architecture.

Introduction to Front-End Web Development

2023

Coursera Inc.

Credential ID: MVRPB4JYSXVS

- **Skills:** Web development fundamentals, UI layout design, and responsive web styling.

Python Basics

2023

Coursera Inc.

Credential ID: FFBVV5PKKC6X

- **Skills:** Core programming concepts, data structures, and functional scripting in Python.

AWARDS & LEADERSHIP EXPERIENCE

Event Lead – DigTech 2026, CECOS University

- Directed an 11-module flagship IT event hosting **500+ participants** across KPK, overseeing cross-functional teams, stakeholder communications, venue logistics, and IT infrastructure including network setups and hardware configurations.

2nd Best Project – Final Year Design Project Open House 2026, CECOS University

- Secured **2nd place** out of 20+ competing projects for engineering excellence with "Aqua Sentinel".

1st Place Winner – ICHARMM 2025 Research Poster Competition, CECOS University

- Awarded **1st place** among 15+ competing teams for presenting the best research poster for the "Aqua Sentinel" project.